Department of Computer Science and Engineering
Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch	**:** III Semester B.E CSE
Course Code & Title	**:** CS3352 & Foundations of Data Science
Name of the Faculty member	**:** Mrs. S. Vijaya Amala Devi, AP / CSE

Innovative Practice Description

- **Unit / Topic: Unit IV/Python Libraries for Data Wrangling /Computation on Array**
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO 8
- **Activity Chosen:** Collaborative Coding
- **Justification:**
    - Collaborative learning is a method of teaching and learning that involves groups of students coming together to solve a problem.
    - The purpose of collaborative learning is to help students learn the complexities of problem-solving and to promote learning motivation.
    - It also helps students make concepts more interesting by involving them to work together to complete problem solution and develop the program code for the practice of using Python libraries for Data wrangling.
- **Time Allotted for the Activity:** 40 Minutes
- **Details of the Implementation:**
    - ❖ After completion of the unit, Computation of array activity can be planned as a team activity, in order to make the students understand the program very clearly and to improve their level of understanding and gain more knowledge in program.
    - ❖ The Student are exposed to deep understanding and practical skills in Python-based data wrangling**.**
    - ❖ The task of the students is creating and developing code for the given problem. The course instructor will provide sufficient hints and correct syntax and will discuss all the doubts in the same class session and make them understand and complete the coding.
    - ❖ The screenshots of the worksheet completed by the student's team Ms. A. Annalakshmi, Ms. S. Malini and Mr.k. Karthikeyan is shown in Figure.1 and Figure.2. The course instructor also provided the correct answers and explained how

the coding works.

- **CO – PO / PSO mapping:**

❖

CO	PO1	PO2	PO3	PO4	PO9	PO10	PO12	PSO1
CO4	**3**	**3**	**3**	**1**	**1**	**1**	**2**	**3**

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative Practice	**PO1**	**PO2**	**PO3**	**PO4**	**PO9**	**PO10**	**PO12**
	2	2	3	3	2	2	2
Justification for correlation	Apply basic Knowledge and fundamentals in Computations on arrays	Identify the need of Python Libraries for Data Wrangling	Able to design and develop the codes for Data Wrangling using Python Libraries	Use the Python Libraries for Data Wrangling and can research on data and provide interpretations and conclusions	Functional individually in identifying the code for Computations on arrays & data wrangling	Communicate / share the ideas with other student's in written and oral presentation	Ability to reproduce the contents gathered through self-learning

- **Images / Screenshot of the practice:**

Figure 1: Student discussed with their team

Figure 2: Student Share their ideas

- **Reflective Critique:**

❖ ***Discussing with students:***

- Students felt a sense of satisfaction as they learned the concepts collectively as a group
- They felt that through this type of learning, Students actively engage and participation encouraged them to share their knowledge with others.

❖ ***Benefit of the practice:***

- Students actively participated in each group.
- It helps when students are placed in teams, encouraging them to share their ideas with others and enhancing their communication skills
- Through this activity, students can gain a deeper understanding of code by utilizing Python libraries for the data wrangling process and learning how to interpret data
- The students are tasked with writing the program code based on their own understanding, primarily utilizing libraries and built-in functions

❖ ***Challenges faced in implementation:***

- Concepts are consistently represented by the top-performing students.
- Some students are encountering challenges in expressing their ideas.

References:

❖ https://www.valamis.com/hub/collaborative-learning
❖ https://teaching.cornell.edu/teaching-resources/engaging-students/collaborativelearning
❖ https://www.evergreen.edu/sites/default/files/facultydevelopment/docs/WhatisCollaborativeLearning.pdf
❖ https://www.ritrjpm.ac.in/images/computer-science/2022-2023/4.KVS_Collaborative_Coding_FDS.pdf

Signature of Faculty Member **HOD**